



Valentin Legras

Data Scientist & Scientific Software Engineer

Basel, Switzerland

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Summary

Data Scientist and scientific software engineer with 7+ years turning large-scale PK/PD and experimental datasets into decision-enabling tools across clinical and preclinical drug development at Roche, Novartis, and Sanofi. Founded and led aNCA, a production-grade open-source R package adopted by 120+ scientists, applying strong software engineering practices (version control, testing, documentation, CI/CD) to build scalable, reusable analytical workflows. Combines machine learning and statistics with R and Python to automate analysis, uncover structure in complex data, and communicate findings to multidisciplinary stakeholders. Delivered more than 2M CHF in annual savings and scales AI/ML and multi-agent coding workflows to accelerate scientific decision making.

Experience

Senior Data Scientist

2020 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Designed and maintained scalable data pipelines and analytical applications that transform large-scale PK/PD and experimental datasets into decision-enabling insights, across multiple therapeutic areas (e.g., Oncology, Neuroscience) from Phase 1 to 3.
- Built and deployed interactive data visualization and exploratory analysis tools in R/Shiny and Spotfire, enabling real-time data review and faster cross-functional decisions on complex experimental data.
- Engineered reproducible data pipelines by converting legacy SAS programs into production R workflows for ADPC, ADNCA, and ADPP datasets, cutting manual effort by 90%; now used by 30 data scientists, saving 70% of their time.
- Led creation of analysis-ready NCA, PK/PD, and CDISC (SDTM/ADaM) datasets from raw data for population modeling in R, Phoenix and NONMEM.
- Scaled multi-agent AI/ML coding workflows across the team (AI-augmented issue creation, automated problem solving, code review), increasing development velocity while upholding software engineering standards.
- Authored responses to health authority queries (FDA, EMA) and managed QC for regulatory submissions, communicating results to diverse scientific and regulatory stakeholders.

aNCA Product Owner

2023 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Directed the end-to-end lifecycle of the 'aNCA' open-source R package under Agile/Scrum, driving it from concept to enterprise-wide adoption by 120+ clinical and pre-clinical scientists, now supporting 2,000+ studies annually.
- Championed broad adoption of a reusable, maintainable data-science tool across teams, managing a 15+ stakeholder network and translating their business needs into a strategically aligned product roadmap.
- Extended the workflow to ingest and harmonize preclinical in vivo and in vitro PK data with protocol metadata, broadening the tool to pre-clinical, DMPK-adjacent use cases.
- Established and enforced production software engineering standards for the package, including version control, testing, documentation, and rigorous code review, to ensure robust, regulatory-grade quality.
- Led, coached, and mentored a core team of 7 developers, strengthening data-science capabilities across the group.
- Delivered more than 2M CHF in annual savings (5 FTEs plus retired commercial licenses) by transforming a manual, weeks-long workflow into a 90% automated process.
- Contributed to the [pharmaverse](#) ecosystem with standardized modules for automated NCA and reporting, built with Git, CI/CD, coding agents (LLMs), and an LLM-based generative-AI interpretation bot in Python.

IT/Data Science Developer

Mar 2020 to Aug 2020

F. Hoffmann-La Roche, Basel, Switzerland

- Translated pharma scientists' requirements into data-science solutions, partnering with wet-lab process modelers and formulators.
- Designed and deployed R/Shiny applications for data visualization and exploration, built on the tidyverse.
- Built, tested, and deployed applications with Git, Jenkins, and Google Cloud Platform (GCP), applying CI/CD practices.
- Automated data management, merging, and reporting workflows in R (officer package), reducing manual effort on recurring deliverables.

Data Scientist

Oct 2019 to Feb 2020

Cofidis, Lille, France

- Developed and evaluated machine learning models in R (Logistic Regression, Gradient Boosting, Random Forest) to predict contract default, applying explainable-AI techniques (feature importance, SHAP) to surface the drivers most predictive of outcome.
- Recommended data-collection strategies from the findings to optimize the rate of returned and completed contracts.

Biostatistical Application Developer *Internship*

May 2019 to Aug 2019

Novartis AG, Basel, Switzerland

- Built and generalized an R/Shiny application to interactively visualize clinical data during DMC (Data Monitoring Committee) reviews, and dynamic R Markdown reports (ggplot, plotly, DT) from CDISC (SDTM/ADaM) datasets.

SAS Developer *Internship*

Jun 2018 to Aug 2018

Sanofi R&D, Paris, France

- Developed and validated SAS (BASE/MACRO/SQL) programs for TLG reporting and built a data-mapping tool generating XML for an AI project, using SDTM/ADaM specifications.

SAS and IT Developer *Internship*

Apr 2017 to Aug 2017

Sanofi R&D, Paris, France

- Migrated and optimized SAS programs for clinical trial data in a new statistical environment, enabling biostatisticians to run their analyses.

Skills

Data Science & Machine Learning

Python, R, machine learning (Logistic Regression, Gradient Boosting, Random Forest), explainable AI (SHAP), generative AI/LLMs, statistics, reproducible workflows, EDA, data mining

Software Engineering

version control (Git), testing, documentation, CI/CD, production-quality code, R package development, SQL, Linux, HPC

Pharmacokinetics & Data

PK/PD modeling, NCA, PBPK/DMPK, NONMEM, Phoenix/WinNonlin, CDISC (SDTM/ADaM), biostatistics, GxP

Applications & Deployment

R/Shiny, data pipelines, GCP, Jenkins, Spotfire/Tableau, ggplot, R Markdown

AI & Automation

Claude (Sonnet/Opus), coding agents (Ona, Claude Code), multi-agent workflows, LLM-assisted development, prompt engineering

Leadership & Delivery

product ownership, Agile/Scrum, team leadership, mentoring, stakeholder management, cross-functional collaboration

Languages

English (Fluent)
French (Native)
Spanish (B2)

Educational Background

Master's Degree, Software Engineering and Statistics

2017 to 2020

Polytech Lille, Villeneuve-d'Ascq, France

Completed alongside internships and an apprenticeship (work-study) at Sanofi, Novartis, Cofidis, and Roche.

Technical Degree, Statistics and Business Intelligence

2015 to 2017

Institut Universitaire de Technologie, Vannes, France

Publications and Presentations

Peer-Reviewed Publications

- [Computer-Aided formulation design for pharmaceutical drug product development, part 01: Materials exploration through a visualization tool](#). Piccione P.M., Lang M.N., et al. (incl. Legras V.) (2024), International Journal of Pharmaceutics, Volume 667, Part B.
- [PBPK Modeling of Entrectinib and Its Active Metabolite to Derive Dose Adjustments in Pediatric Populations Co-Administered with CYP3A4 Inhibitors](#). Umehara K., Parrott N., Schindler E., Legras V., Meneses-Lorente G. (2024), Clin Pharmacol Ther, 116: 1130-1140.
- [Dexamethasone is associated with reduced frequency and intensity of cytokine release syndrome compared with alternative corticosteroid regimens as premedication for glofitamab in patients with relapsed/refractory large B-cell lymphoma](#). Haematologica (2024).
- Sunburst visualization method contributed to [Hipp J.F. et al., The UBE3A-ATS antisense oligonucleotide rugonersen in children with Angelman syndrome: a phase 1 trial](#). Nat Med 2025; 31(9):2936-2945.

Conference Presentations

- [Revolutionizing Data Exploration: An Interactive Journey through R Shiny](#). Legras V., PHUSE 2023.
- [R Scripts to Streamline Non-Compartmental Analysis \(NCA\)](#). Legras V., Suekuer E., PHUSE 2024 (aNCA open-source).
- [Sunburst Plots for Analysing the Impact of Drugs on Cytokine Release Syndrome \(CRS\) Management](#). Legras V., PAGE 2024.
- [Automated QC Framework for Reliable ADNCA Dataset Generation with admiral](#). Legras V., Aspridis L., PHUSE 2025.

Hobbies

Roller skating, Ice skating, Surfing, Basketball, Cycling, Drawing, Coding ([stock market analysis](#)).